

Yongye(Felix) Hu

☎ 510-277-7666 | ✉ hyy.felix@gmail.com | 🌐 felix-hyy | 👤 felix-hu.space

EDUCATION

University of California, Berkeley

Master of Engineering | Robotics Engineering

Aug 2024 - May 2025

Berkeley, CA

Wuhan University of Science and Technology

Bachelor of Engineering (Honours) | Mechatronics Engineering

Sep 2020 - May 2024

Wuhan, China

PROFESSIONAL EXPERIENCE

L.I.F.T: A Hybrid Flying and Driving Robot Platform

Sep 2024 - May 2025

Robotics Engineer & Project Lead

Berkeley, CA

- Developed a ROS2-based robotic system in Python, integrating a simulator and MPC controller for precise obstacle avoidance and full trajectory completion with real-time state logging.
- Integrated OptiTrack motion capture with the robotic platform, achieving <1 mm position accuracy and <10 ms latency for closed-loop pose control, enabling sub-centimeter trajectory tracking and iterative controller refinement.

ROBOCON: Multi-Robot Coordination System

Sep 2022 - Jul 2023

Integrated Mechanical Engineer

Wuhan, China

- Developed embedded C firmware for STM32F407, enabling CAN-based motor control, BLE remote communication, and PID position control with ± 2 mm tracking error for an omnidirectional mobile robot.
- Developed and tuned a dual-mode chassis control architecture with three PID loops managing translation and orientation, achieving <1.2 s error convergence and 40% improvement in directional stability.
- Implemented a custom UART protocol with headers, footers, and float unpacking; leveraged DMA + interrupts for non-blocking 100 Hz updates of 6-DoF data, with <0.2% error rate.
- Designed a modular PID-based navigation system on STM32 with inverse kinematics and yaw compensation, achieving omnidirectional motion and $\pm 0.1\%$ global localization accuracy.

Lightweight, Sustainable Remote-Mining Excavator System

Jan 2021 - Jul 2023

R&D Engineer

Wuhan, China

- Independently developed an STM32-based control system using fuzzy PID algorithms for closed-loop step-per motor control, achieving ± 1 mm positioning accuracy.
- Consolidated five functional boards into a single 4-layer PCB with KiCad, integrating control, power, and communication modules; reduced overall footprint by 40% and verified signal integrity via 100-MHz oscilloscope.
- Designed a compact, low-power crane system with ESP32-based Wi-Fi remote control, enabling low-latency communication and autonomous coordination across multiple devices, achieving over 95% task repeatability.

WORK EXPERIENCE

Wuhan Xiaoxian AI Technology Co., Ltd.

Sep 2023 - Mar 2024

Mechatronics Engineer Intern

Wuhan, China

- Developed control programs for a six-axis CRP2 robotic arm using embedded C and ladder logic, enabling automated multi-axis motion for fuel rod testing.
- Designed an STM32F103-based gateway to enable Modbus RTU communication between the PC and robotic controller, ensuring stable and real-time data exchange.

SKILLS

Controls & Robotics: PID/Fuzzy PID, MPC, Closed-Loop Control, Sensor Fusion, ROS2, Inverse Kinematics

Embedded Systems: STM32, ESP32, Arduino, Raspberry Pi, Embedded C/C++, PCB Design (KiCad), Motor Control

Communication Protocols: CAN, UART, SPI, I2C, Modbus RTU, BLE, Wi-Fi, DMA

Programming: Python, C++, Linux, MATLAB, Qt, LaTeX

Design & Prototyping: SolidWorks, Autodesk Fusion, Ansys FEA/CFD, FDM/SLA 3D Printing, CNC Machining